

Convex optimization — Exercise sheet II

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Exercise 1

Write the transportation problem presented in slides 14 to 18 (section 1.1) of the introduction as a linear program in canonical form, then in standard form.

Exercise 2

Solve the linear program obtained after modeling in exercise 1.2 of the previous exercise sheet (maximizing the revenue of a candy manufacturer).

Exercise 3

An industrial laundry has two professional washing machines. These two machines have a washing cycle of 1 hour. The first machine (referred to as A) is capable of washing a maximum of 26 kg of laundry per cycle, while the other machine is capable of washing a maximum of 32 kg of laundry per cycle.

The company estimated, taking into account maintenance costs and charges, that the profits made was roughly equal for machine A to 24 cents per kilogram of washed laundry, and for machine B to 15 cents per kilogram of washed laundry.

The electric energy consumption is equal for machines A to 600 kJ per kilogram washed, and for machine B to 1000 kJ per kilogram washed. Water consumption is equal for machine A to 4 liters per kilogram washed, and for machine B to 2 liters per kilogram washed.

Given the state of the distribution network and the current water and electricity use of infrastructure, it is possible to consume a maximum of 10 kWh per cycle and 128 liters per cycle for the operation of the two washing machines.

The objective of this company is to determine the load of each machine per cycle in order to maximize its profit.

1. Model this linear optimization problem.

2. Write this problem as a linear program in canonical form, then in standard form.
3. Solve this program, and check that the optimal profit is equal to 840 cents per cycle.

Exercise 4

Solve the following linear program:

$$\begin{array}{ll}
 \underset{(x_1, x_2, x_3) \in \mathbb{R}^3}{\text{maximize}} & 5x_1 + 5x_2 + 3x_3 \\
 \text{subject to} & 4x_1 + 3x_2 - 2x_3 \leq 30 \\
 & x_1 - 3x_2 + 2x_3 \leq 15 \\
 & 3x_1 + 6x_2 + x_3 \leq 60 \\
 & x_1 \geq 0, x_2 \geq 0, x_3 \geq 0.
 \end{array}$$